

CURRICULUM VITAE

Dr. Rahul Kumar

Udaipur, Rajasthan

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OBJECTIVE:

Accomplished academician and active researcher with over 11 years of teaching experience, seeking an Associate Professor position to contribute through high-impact teaching, curriculum innovation, and meaningful research while fostering a dynamic, student-centered learning environment.

RESEARCH PROFILE IDs:

➤ **Orchid ID:**

ID: 0009-0006-9020-9095

Link: <https://orcid.org/0009-0006-9020-9095>

➤ **Google Scholar Profile:**

ID:- W_TENYIAAAAJ

Link: https://scholar.google.com/citations?user=W_TENYIAAAAJ&hl=en

➤ **Scopus Profile:**

ID: 58541009000

Link: <https://www.scopus.com/authid/detail.uri?authorId=58541009000>

➤ **Web of Science:**

ID: LKK-7522-2024

Link: <https://www.webofscience.com/wos/author/record/LKK-7522-2024>

➤ **Vidan ID:**

ID:- 573311

LINK:- <https://vidwan.inflibnet.ac.in/profile/573311>

➤ **Sementic Id:**

ID: - 2275941219

LINK: - <https://www.semanticscholar.org/author/Rahul-Kumar/2275941219>

Research gate:-

Link:- https://www.researchgate.net/profile/Rahul-Kumar-859?ev=hdr_xprf

EXPERIENCE:

- Currently serving as an **Associate Professor** in the Faculty of Computing and Informatics at Sir Padampat Singhania University since **06 June 2025** to present.
- Served as an **Associate Professor** in the Department of Computer Science and Engineering at Amity University from **04 December 2023 to 04 June 2025**.
- Worked as an **Assistant Professor** in the Department of Computer Science and Engineering at Koneru Lakshmaiah Education Foundation (KL University) from **09 May 2023 to 02 December 2023**.

- Served as an **Assistant Professor & Deputy Controller of Examinations (Deputy COE)** in the Department of Computer Science and Engineering at Dr. K.N. Modi University from **26 December 2018 to 03 May 2023**.
- Worked as an **Assistant Professor** in the Department of Computer Science and Engineering at NGF College of Engineering and Technology from **05 August 2017 to 25 December 2018**.
- Served as an **Assistant Professor** in the Department of Computer Science and Engineering at Shekhawati Institute of Engineering and Technology from **05 July 2014 to 03 August 2017**.

TECHNICAL QUALIFICATION:

- Completed **Ph.D. in Computer Science and Engineering** on the topic “*Optimizing Fuzzy Rough Feature Selection Using Threshold Values*” from the Department of Computer Science at Dr. K.N. Modi University (**2019–2022**).
- Currently pursuing **Post-Doctoral Fellowship (PDF) in Computer Science and Engineering** on the topic “*Privacy-Preserving Feature Selection and Extraction for Federated Machine Learning in Real-World Applications*” from the Research Institute of IoT Cybersecurity, Department of Electronic Engineering at National Kaohsiung University of Science and Technology (**Expected completion: February 2026**).
- Completed **M.Tech in Computer Science and Engineering** with **82.3%** from Suresh Gyan Vihar University (**2012–2014**).
- Completed **B.Tech in Computer Science and Engineering** with **70.07%** from ICFAI University (**2008–2012**).

ACADEMIC QUALIFICATION:

- Completed **Intermediate (Higher Secondary)** with **First Division** from Bihar School Examination Board in **2007**.
- Completed Matriculation (Secondary School) with **First Division** from **Bihar School Examination Board** in **2005**.

AREAS OF INTEREST

- Database Management System
- Machine Learning, Federated Learning
- Computer Network

ADDITIONAL ROLES AND RESPONSIBILITIES

- Served as **Deputy Controller of Examinations** at Dr. K.N. Modi University from **December 2019 to 03 May 2023**, overseeing examination processes and academic administration.
- Worked as a **Ph.D. Co-Coordinator** in the Amity School of Engineering (Department of Computer Science and Engineering) at Amity University.
- Acted as an **External Examiner** for Ph.D. Viva-Voce (thesis defense) at Dr. K.N. Modi University.
- Served as an **External Reviewer** for Ph.D. theses and synopses for various universities.
- Contributed as an **Organizing Committee Member** for the establishment of the IEEE Student Chapter at the Amity School of Engineering, CSE Department, Amity University.
- Acted as a **Reviewer** for reputed journals and platforms, including IEEE Xplore and journals indexed in Web of Science.
- Currently serving as a **Project Coordinator (FCI)** at Sir Padampat Singhanian University.

RESEARCH/CONFERENCE PAPER/PATENTS:

JOURNAL PAPERS:-

1. Kumar, Rahul, Ashok Kumar, Karamjit Bhatia, Kottakkaran Sooppy Nisar, Siddharth Singh Chouhan, Priti Maratha, and Anoop Kumar Tiwari, **“Hybrid Approach of Cotton Disease Detection for Enhanced Crop Health and Yield,”** *IEEE Access*, 2024. DOI: 10.1109/ACCESS.2024.3419906 (*SCI-Q1*).
2. Kumar, Ashok, Saurabh Singh, Mahadev, and Rahul Kumar, **“An Intelligent Human-Centric System to Diagnose Breast Cancer Using Machine Learning and Optimized Feature Selection Techniques,”** *Transactions on Emerging Telecommunications Technologies*, vol. 35, no. 4, 2024, e4913 (*SCI-Q2*).
3. Gupta, Atul Kumar, Rahul Kumar, Jhankar Moolchandani, Vikas Thada, Mohd Asif Shah, and Anoop Kumar Tiwari, **“Distance-Based Unsupervised Local Outlier Detection: Based Values Analysis to Improve Outlier Detection Using Machine Learning,”** *IET Communications*, vol. 19, no. 1, 2025, e70060 (*SCI-Q2*).
4. Kumar, A., Dhamija, D., Chole, V., Moolchandani, J., Kumar, R., and Garg, U., **“Prediction and Segmentation of Heart Disease Using Boosting-Based Machine Learning Algorithms,”** *Journal of Neonatal Surgery*, 2025 (*Scopus-Q4*).
5. Kumar, S., Kumar, A., Parashar, N., Moolchandani, J., Saini, A., Kumar, R., et al., **“An Optimal Filter Selection on Grey Scale Image for De-Noising Using Fuzzy Technique,”** *International Journal of Intelligent Systems and Applications in Engineering*, 2024 (*Scopus-Q4*).
6. Singh, Vinay Kumar, Raj Sinha, Umang Garg, Rahul Kumar, Bhimasen Moharana, and Parul Goyal, **“MHID: Malware Detection Using Hybrid Honeypot and Intrusion Detection System,”** *Communications on Applied Nonlinear Analysis*, vol. 31, no. 2s, 2024 (*Scopus-Q4*).
7. Balaji, P., Haritha, D., Sumalatha, K., Priya, D. U., Bhutia, R., and Kumar, R., **“Emotions-Based Sentiment Analysis Using Fusion-Based Learning Model,”** *International Journal of Engineering Trends and Technology*, vol. 71, no. 6, 2023 (*Scopus-Q4*).

8. Harit, Vibhor, Gaurav Gulati, Raj Sinha, Ashok Kumar, Penubaka Balaji, and Rahul Kumar, **"IoT-Enabled Weather Analysis and Monitoring System with Intelligent Climate Sense System,"** *Cuestiones de Fisioterapia*, vol. 54, no. 2, 2025 (*Scopus-Q4*).
9. Kumar, R., and Gupta, K. S., **"Optimizing Fuzzy Rough Feature Selection Using Threshold Values,"** *Journal of the Asiatic Society of Mumbai*, vol. XCV (4), 2022 (*UGC CARE*).
10. Kumar, R., and Pathak, S., **"A Survey on Medical Imaging through Deep Learning Techniques,"** *Aegaeum Journal*, vol. 7, no. 10, 2019.
11. Kumar, Rahul, Dinesh Goyal, and Rakesh Kumar, **"Real-Time Toll Rate Determination Using Image Processing and Network Database,"** *International Journal of Digital Applications and Contemporary Research*, vol. 2, no. 9, 2014.
12. Patel, Vipul, Rahul Kumar, and Ankur Raj, **"Improving Security and Integrity of Data Storage in Cloud Computing Using Homomorphic Authentication Technique,"** *International Journal of Innovations in Engineering and Technology*, vol. 3, no. 2, 2013.
13. Kumar, A., Kumar, R., and Bhatia, K., **"Offline Writer-Independent Handwritten Signature Verification System,"** *Engineering and Technology Journal for Research and Innovation*, vol. 6, no. 2, 2024.
14. Kumar, R., Kumar, A., and Bhatia, K., **"Review of Language Models for Conversational AI in ChatGPT,"** *Engineering and Technology Journal for Research and Innovation*, vol. 6, no. 2, 2024.
15. Sharma, R. C., Goyal, D., and Kumar, R., **"Real-Time Data Acquisition and Control System Using Image Processing,"** *International Journal of Digital Applications & Contemporary Research*, vol. 2, no. 12, 2014.
16. Chouhan, P. S., Yadav, G. C., and Kumar, R., **"Smart Toll Rate Determination in Real-Time Systems,"** *International Journal of Digital Applications & Contemporary Research*, vol. 4, no. 6, 2016.

CONFERENCE PAPERS:-

1. Rahul Kumar, Chin-Shiuh Shieh, Prasun Chakrabarti, Ashok Kumar, Jhankar Moolchandani, and Raj Sinha, **"Privacy-Preserving Federated Learning in Healthcare, E-Commerce, and Finance: A Taxonomy of Security Threats and Mitigation Strategies,"** presented at the *First International Conference on Engineering and Technology for a Sustainable Future (ICETSF-2025)*, Amravati, Maharashtra, India (*Scopus Indexed*).
2. Rahul Kumar, A. Kumar, J. K. Chaudhary, Mahadev, A. Kumar, and V. Thada, **"AI-Assisted Resource Allocation in 5G/6G Networks,"** presented at *OPTIMA 2025: Optical Communication, Photonics, Telecommunications, and Intelligent Machine Applications*, Tashkent, Uzbekistan, pp. 443–448 (*IEEE Conference*).
3. J. K. Chaudhary, Mahadev, A. Kumar, Rahul Kumar, and A. Kumar, **"Photonic-Enhanced Fiber-Optic Sensor Networks for Large-Scale IoT Deployments,"** *OPTIMA 2025*, Tashkent, Uzbekistan, pp. 436–442 (*IEEE Conference*).
4. A. Kumar, J. K. Chaudhary, Mahadev, A. Kumar, and Rahul Kumar, **"Photonics-Integrated Edge and Cloud Frameworks for IoT Scalability,"** *OPTIMA 2025*, Tashkent, Uzbekistan, pp. 637–643 (*IEEE Conference*).
5. Rahul Kumar, Jhankar Moolchandani, Anshita Shukla, Santosh Sahu, Vikas Thada, and Vikrant Chole, **"Machine Learning-Based Prediction of Gold Prices Using Economic Indicators,"** in

SMART 2024: 13th International Conference on System Modeling & Advancement in Research Trends, IEEE, pp. 520–524 (*Scopus Indexed*).

6. Vikrant Chole, Santosh Sahu, Jhankar Moolchandani, Ashok Kumar, Rahul Kumar, and Anshita Shukla, “**Analyzing the Impact of AI Adoption on Job Market Dynamics and Employment Trends Using a Synthetic Dataset**,” in *SMART 2024*, IEEE, pp. 525–531 (*Scopus Indexed*).
7. Anshita Shukla, Rahul Kumar, Vikrant Chole, Jhankar Moolchandani, Santosh Sahu, and Ashok Kumar, “**Comparative Analysis of Machine Learning Models for Diabetes Prediction**,” in *SMART 2024 / ICTACS 2024*, IEEE, pp. 869–874 (*Scopus Indexed*).
8. Jhankar Moolchandani, Vikrant Chole, Santosh Sahu, Rahul Kumar, Anshita Shukla, and Ashok Kumar, “**Predictive Analytics in Sports: Forecasting Outcomes and Medal Tally Trends at the 2024 Summer Olympics**,” in *ICTACS 2024: International Conference on Technological Advancements in Computational Sciences*, IEEE, pp. 1987–1992 (*Scopus Indexed*).
9. Sahu, Santosh, Ashok Kumar, Rahul Kumar, Anshita Shukla, Vikrant Chole, and Jhankar Moolchandani, “**Predicting Student Academic Performance Using Machine Learning: Analysis of Socio-Economic and Personal Factors from Secondary Education in Portugal**,” in *4th International Conference on Technological Advancements in Computational Sciences (ICTACS 2024)*, IEEE, pp. 1982–1986, 2024. DOI: 10.1109/ICTACS62700.2024.10841127 (*Scopus Indexed*).
10. Moolchandani, Jhankar, Rahul Kumar, and Kulvinder Singh, “**Advancements in Handwritten English Character Recognition: A Comprehensive Analysis Using Pattern Recognition and Deep Learning on Scanned Documents**,” in *Congress on Control, Robotics, and Mechatronics*, Springer Nature Singapore, pp. 319–331, 2024 (*Scopus Indexed*).
11. Rahul Kumar, A. Kumar, and K. Bhatia, “**Review of Language Models for Conversational AI in ChatGPT**,” in *Proceedings of the National Conference on Advancing Engineering Frontiers: Innovations, Challenges and Solutions*, Amity University, RISER, 10 June 2024.
12. A. Kumar, Rahul Kumar, and K. Bhatia, “**Offline Writer-Independent Handwritten Signature Verification System**,” in *Proceedings of the National Conference on Advancing Engineering Frontiers: Innovations, Challenges and Solutions*, Amity University, RISER, 10 June 2024.
13. Rahul Kumar, Jhankar Moolchandani, Ashok Kumar, and Umang Garg, “**Feature Selection Optimization Using Hybrid Fuzzy-Rough Set Techniques**,” in *Data Engineering and Applications: Proceedings of IDEA 2K24*, Springer (Lecture Notes in Electrical Engineering), 2024 (*Scopus Indexed*).
14. Rahul Kumar et al., “**ChatGPT: A Review of Language Models for Conversational AI**,” in *AI-DSP 2023: International Conference on Emerging Trends in Artificial Intelligence, Data Science and Signal Processing*, 2023 (*Scopus Indexed*).
15. Rahul Kumar et al., “**Exploring the Metaverse: A Comprehensive Review and Future Perspectives**,” in *AI-DSP 2023*, 2023 (*Scopus Indexed*).
16. Kumar, R., and Gupta, K. S., “**A Survey on Attribute Reduction Techniques Based on Fuzzy Rough Sets**,” in *Proceedings of IJEMMASSS*, vol. 4, pp. 13–26, 2022.
17. Kumari, Perna, Aneesh Kumar Mishra, and Rahul Kumar, “**Web User Access Pattern Identification Using Clustering Algorithms: A Survey**,” in *10th IEEE International Conference on Communication Systems and Network Technologies (CSNT 2021)*, IEEE (*Scopus Indexed*).
18. Kumar, R., and Pathak, S., “**Analyzing and Measuring Performance for Fuzzy-Rough Set Feature Selection**,” in *Proceedings of IJEMMASSS*, vol. 3, pp. 98–110, 2021.

19. Rahul Kumar, A. K. Mishra, and D. K. Singh, **“Packet Loss Avoidance in Mobile Adhoc Networks Using Trusted LDoS Techniques,”** in *2nd International Conference on Data, Engineering and Applications (IDEA 2020)*, IEEE, pp. 1–5, 2020 (*Scopus Indexed*).
20. Rahul Kumar, D. K. Singh, and A. K. Mishra, **“An Approach to Enhance Computed Tomography Images for Extracting Fine Details,”** in *IDEA 2020*, IEEE, pp. 1–6, 2020 (*Scopus Indexed*).

Accepted (Under Publication)

1. Rahul Kumar, J. K. Chaudhary, Mahadev, R. Sinha, and A. Kumar, **“Federated Learning for Privacy-Preserving Healthcare Analytics: A Framework for Multi-Hospital Collaboration,”** *International Journal of Intelligent Systems and Applications (IJISA)*, accepted on **10 February 2026** (*Scopus Indexed*).
2. R. Sinha, Rahul Kumar, A. Kumar, J. K. Chaudhary, and Mahadev, **“Lightweight Federated NLP: A TF-IDF and FedProx Powered Model for Privacy-First Text Classification,”** *International Journal of Modern Education and Computer Science (IJMECS)*, accepted on **12 February 2026** (*Scopus Indexed*).
3. M. Dev, A. Kumar, Rahul Kumar, A. Kumar, and J. Chaudhary, **“Optimizing Deep Learning Models for Real-Time Edge Computing Applications,”** in *Proceedings of the International Conference on Intelligent and Innovative Practices in Engineering & Management (ICIPEM)*, **2025** (IEEE Conference).
4. A. Kumar, Rahul Kumar, A. Kumar, J. K. Chaudhary, and M. Mahadev, **“Intelligent Optimization of Deep Learning Models for Real-Time Decision Systems,”** in *Proceedings of the International Conference on Intelligent and Innovative Practices in Engineering & Management (ICIPEM)*, **2025** (IEEE Conference).
5. Rahul Kumar, A. Kumar, J. Moolchandani, A. Sen, R. Sinha, and C. Goswami, **“Optimizing Battery Energy Arbitrage: A Deep Reinforcement Learning Approach with Hybrid CNN-LSTM Price Forecasting,”** in *Proceedings of the International Conference on Computational Intelligence and Autonomous Systems (ICIAS)*, **2025** (Taylor & Francis Conference).

Under Review

1. A. Kumar, R. Sinha, J. K. Chaudhary, Mahadev, and R. Kumar, **“A Lightweight Attention-Driven Deep Learning Framework for Text-Based Psychological Stress Detection,”** *International Journal of Information Engineering and Electronic Business (IJIEEB)*, submitted on **06 January 2026** (*Scopus Indexed*).
2. R. Kumar, C.-S. Shieh, and P. Chakrabarti, **“A Multi-Domain Federated Learning Architecture with Secure Feature Optimization,”** *International Journal of Intelligent Systems and Applications (IJISA)*, submitted on **04 February 2026** (*Scopus Indexed*).
3. M. Mahadev, A. Kumar, R. Sinha, R. Kumar, and J. K. Chaudhary, **“A Hybrid Attention-Based CNN-BiLSTM and Random Forest Ensemble for Highly Accurate Ovarian Cancer Detection Using Serum Biomarkers,”** *Pattern Recognition and Image Analysis: Advances in Mathematical Theory and Applications* (*Scopus Indexed*).

PATENT: -

1. **“An Apparatus for Device-Based Skin Lesion Analysis Using Ensemble Learning,”** Indian

Patent Application, **Application No.: 202511133315 A**, filed on **30 December 2025** and published on **06 February 2026** in the *Patent Office Journal, India*.

2. Dr. Rahul Kumar (2024), **“Object Classification for Remotely Sensed Images Using Enhanced CNN Model”**, *Patent Office Journal, India*, **Application No.: 202411041144**, published on **07 June 2024**.
3. Dr. Rahul Kumar (2024), **“An Integrated Solution for Smart Agriculture Monitoring System”**, *Patent Office Journal, India*, **Application No.: 202411053627**, published on **26 July 2024**.
4. Dr. Ashok Kumar, Dr. Rahul Kumar, Dr. Mahadev, Dr. Jitendra Kumar Chaudhary, Rudramani Bhutia, Dr. Penubaka Balaji (2023), **“Instrument Used for Detection of Diabetic Neuropathy”**, *UK Design Registration*, **Design No.: 6306542**, granted on **08 September 2023**.

COPYRIGHTS:-

1. Dr. Mamta Kumari, Dr. Raj Sinha, Dr. Meena Tiwari, Dr. Rahul Kumar, Md. Gauhar Hasnain, and Dr. Neshat Qamar, **“Combating Greenwashing with AI: Algorithmic Detection of Unsustainable Marketing Practices in Augmented Reality and Virtual Reality,”** **Application No.: 21236/2024-CO/L**, filed/published on **04 July 2024**.

ACHIEVEMENT

- Successfully completed **NPTEL Online Certification in Blockchain Architecture, Design, and Use Cases** conducted by Indian Institute of Technology Kharagpur in **November 2018**.
- Successfully completed **NPTEL Online Certification in Database Management Systems** conducted by Indian Institute of Technology Kharagpur on **02 May 2018**.
- Published an article titled **“Real-Time Toll Rate Determination Using Image Processing and Network Database”** in a leading daily newspaper on **12 May 2014**.

TECHNICAL SKILLS

- Know about Database management system software like Oracle, MySQL
- Know about operating system like Windows XP/7/Vista/8/10/11
- Know about Python Programming

PERSONAL DETAILS

Father's Name: Ram Naresh Singh

Mother's Name: Phool Badan Devi

Permanent Address: At. Post: - Ghataro, Tq:-Lalganj, Dist.: - Vaishali, Bihar, Pin-844119

Date of Birth: 10th-December-1990

Marital Status: Married

Nationality: Indian

Language Known: English, Hindi

DECLARATION:

I hereby declare that the above information is true to best of my knowledge.

Place: Udaipur, Rajasthan

Dr. Rahul Kumar